Jyotirmoy Bandyopadhayaya

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Summary

Software engineer focused on backend systems, identity, and developer platforms. More than four years shipping production systems in Go, Rust, and TypeScript. Primary role is enterprise customer identity and access management (CIAM) at LoginRadius. EnvSync and Engineer's Cradle were concurrent with LoginRadius, not sequential roles.

Author of IETF drafts on OAuth push-based authenticator discovery and Client Initiated Backchannel Authentication (CIBA) binding. Built oauth-impl, a self-hosted OAuth 2.0 and OpenID Connect issuer for authentication, authorization, and single sign-on (SSO), along with Chabi, a Rust Redis-compatible server, and x-log, an ActivityPub microblog.

On systems owned end to end, this work cut AWS cost by 76%, lowered P95 latency by about 30%, and raised deploy frequency by about 10x.

Experience

Software Engineer, LoginRadius — July 2024 – Present

- Product and R&D: ship agentic IAM, MCP authentication, Demonstrating Proof of Possession (DPoP), Client Initiated Backchannel Authentication (CIBA), and B2B IAM on LoginRadius customer identity and access management (CIAM), including enterprise single sign-on (SSO), authentication, and authorization.
- Author of IETF drafts on OAuth push-based authenticator discovery and lifecycle, and on CIBA binding.
- Led enterprise CIAM integrations and wrote SDKs, docs, and reference apps that unblocked customer engineering teams.
- Own APIs, SDKs, reference apps, and proofs of concept in TypeScript and Go, with Docker, DataDog, and GitLab CI.

Founder & Product Architect (concurrent), EnvSync Cloud (envsync.cloud) — Jan 2025 – Present

- Multi-tenant secrets platform: Bun and Hono API, Go CLI, React dashboard, and TypeScript and Go SDKs. Dual-license open source and enterprise monorepo on Turborepo and Bun.
- miniKMS in Go: per-tenant envelope encryption (HKDF, AES-256-GCM), a 3-level public key infrastructure (PKI), and hash-chained audit over gRPC.
- Enterprise edition: OpenID Connect (OIDC) and SAML single sign-on, OpenFGA authorization, secret rotation, dynamic secrets, and GitHub Actions pull.

Software Engineer (Founding Engineer, concurrent), Engineer's Cradle — July 2024 – May 2025

- Founding engineer. Set CI/CD and quality standards and cut onboarding time by about 40%.
- Built Rust machine-to-machine authentication with 99.9% uptime, a Go GraphQL gateway that lowered P95 latency by about 30%, and GitHub Actions pipelines that raised deploy frequency by about 10x.

Backend Developer, SaaSDen — June 2023 – Nov 2023

- Built SaaS product and IT integrations in Node.js, including identity and access management (IAM), APIs, and analytics.
- Built deployment workflows for the product.

Site Reliability Engineer, SharpSell — Feb 2023 – Apr 2023

- Administered AWS (EC2, ECS, ECR, IAM). Moved workloads from EC2 to ECS and cut monthly AWS cost by 76% ($19K to $4.5K), with blue-green deploys.
- Built deployment pipelines with AWS CodeBuild, CodePipeline, and GitHub Actions.

Software Engineer Intern, Hybr1d — July 2022 – Dec 2022

- Developed and maintained the backend of the Hybr1d platform.
- Migrated the backend from Ruby on Rails to TypeScript.
- Built REST APIs with Node.js and Express, JWT authentication, PostgreSQL, MongoDB, Hasura, and GraphQL.

Projects

Oauth Implementation Server June 2026 – Present

- Self-hosted OAuth 2.0 and OpenID Connect (OIDC) issuer in Go. Supports authorization code with Proof Key for Code Exchange (PKCE), device code, Client Initiated Backchannel Authentication (CIBA), pushed authorization requests (PAR), Demonstrating Proof of Possession (DPoP), mutual TLS (mTLS), token exchange, and dynamic client registration.

x-log Nov 2025 – Present

- x-log is an open-source, Bun + TypeScript powered blog platform that federates with the Fediverse using ActivityPub. Readers on Mastodon, Elk, Soapbox, and compatible clients can search, read, like, and follow x-log profiles and posts directly from their clients.

Chabi Sept 2025 – May 2026

- Rust Redis-compatible server using the Redis Serialization Protocol (RESP), with snapshot persistence.
- On a 4-vCPU benchmark: 35.6k ops/s versus Redis at 12.3k and DiceDB at 15.7k (about 2.9x Redis). GET p50 was 0.07 ms versus Redis at 0.27 ms.

Publications

OAuth Push-Device Discovery & Lifecycle July 2026
J. Bandyopadhayaya
(IETF Internet-Drafts)

CIBA Binding July 2026
J. Bandyopadhayaya
(IETF Internet-Drafts)

Skills

Languages: Go, Rust, TypeScript, Python, Bash

Systems: Microservices, REST, gRPC, GraphQL, PostgreSQL, SQL, Redis, Bun, Hono, Docker, Turborepo, GitHub Actions

Identity & Security: OAuth 2.0, OpenID Connect (OIDC), SAML, single sign-on (SSO), authentication, authorization, CIBA, customer identity and access management (CIAM), IAM, OpenFGA, envelope encryption, key management (KMS), DevSecOps

Cloud & Ops: AWS (ECS, EC2, S3, IAM, ECR, CodePipeline, CloudWatch), Cloudflare, Keycloak, DataDog, Prometheus, Grafana

Education

Lovely Professional University, Bachelor of Technology (B.Tech) in Computer Science and Engineering (Cybersecurity), GPA 7.8/10 June 2020 – June 2024

Certifications

Certified Ethical Hacker (CEH) v12 by EC-Council Apr 2024

Certified Penetration Testing Specialist (CPTS) by HackTheBox July 2024